WORKED SOLUTION

STEP I 1998, Question 3

Pure • True or False Proofs

STEP I 1998, Question 3

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/98-s1-q3

The key idea

Each part needs a different tool, and the first job is to decide which way it goes. A "true" needs a proof that covers every case; a "false" needs just one counterexample, or an argument that no object of the required kind can exist. Taking logarithms settles (i), a single well-chosen value of θ settles (ii), the behaviour of polynomials for large θ settles (iii), and a perfect square settles (iv).

Solution

Part (i): true

Since $a > 0$, we can write $a = \mathrm{e}^{\ln a}$. Then

$$a^{\ln b} = \left(\mathrm{e}^{\ln a}\right)^{\ln b} = \mathrm{e}^{\ln a \ln b},$$
$$b^{\ln a} = \left(\mathrm{e}^{\ln b}\right)^{\ln a} = \mathrm{e}^{\ln b \ln a}.$$

The exponents are equal because multiplication commutes, so $a^{\ln b} = b^{\ln a}$ for all $a, b > 0$. (Equivalently: both sides are positive and have the same logarithm, $\ln a \ln b$.)

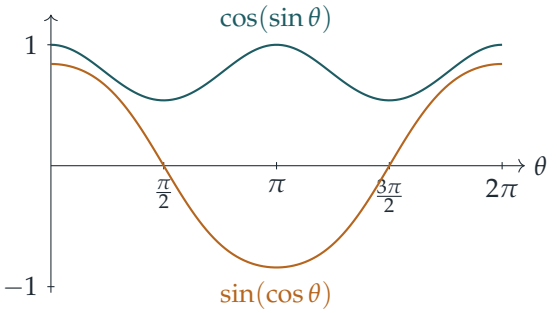
Part (ii): false

One counterexample is enough. Take $\theta = 0$:

$$\cos(\sin 0) = \cos 0 = 1, \qquad \sin(\cos 0) = \sin 1.$$

Since $0 < 1 < \frac{\pi}{2}$, we have $0 < \sin 1 < 1$ (in fact $\sin 1 \approx 0.841$), so the two sides differ and the statement is false.

In fact the two sides are *never* equal; the graph shows $\cos(\sin\theta)$ always above $\sin(\cos\theta)$.



To prove this (not needed for the question), note $\sin(\cos\theta) \leq \sin|\cos\theta| = \cos\left(\frac{\pi}{2} - |\cos\theta|\right)$ and $\cos(\sin\theta) = \cos|\sin\theta|$. Both $|\sin\theta|$ and $\frac{\pi}{2} - |\cos\theta|$ lie in $[0, \frac{\pi}{2}]$, where cosine is decreasing, and $|\sin\theta| < \frac{\pi}{2} - |\cos\theta|$ because $|\sin\theta| + |\cos\theta| \leq \sqrt{2} < \frac{\pi}{2}$. Hence $\cos(\sin\theta) > \sin(\cos\theta)$ for every θ.

Part (iii): false

Suppose P is any polynomial. We show that $|\mathrm{P}(\theta) - \cos\theta| > 10^{-6}$ for some real θ, splitting into two cases.

P *constant*, say $\mathrm{P}(\theta) = c$. Using $\theta = 0$ and $\theta = \pi$, where $\cos\theta$ is 1 and -1, the triangle inequality gives

$$|c-1| + |c+1| \geq |(c+1) - (c-1)| = 2,$$

so at least one of $|c-1|$, $|c+1|$ is at least 1, which is far bigger than 10^{-6}.

P *of degree* $n \geq 1$, say $\mathrm{P}(\theta) = a_n\theta^n + \cdots + a_0$ with $a_n \neq 0$. For $\theta \neq 0$,

$$\mathrm{P}(\theta) = a_n\theta^n\left(1 + \frac{a_{n-1}}{a_n\theta} + \cdots + \frac{a_0}{a_n\theta^n}\right),$$

and the bracket tends to 1 as $\theta \to \infty$, while $|a_n\theta^n| \to \infty$. So $|\mathrm{P}(\theta)| \to \infty$, and we can choose θ with $|\mathrm{P}(\theta)| > 2$. For that θ, since $|\cos\theta| \leq 1$,

$$|\mathrm{P}(\theta) - \cos\theta| \geq |\mathrm{P}(\theta)| - |\cos\theta| > 2 - 1 = 1 > 10^{-6}.$$

In both cases the inequality fails somewhere, so no such polynomial exists.

Ask yourself But the Maclaurin series of $\cos\theta$ is a polynomial-like sum that gets as close as we like. Why doesn't a truncation work?

A truncated Maclaurin series *is* within 10^{-6} of $\cos\theta$ on any fixed bounded interval, provided we take enough terms. But the statement asks for one polynomial that works for *all real* θ, and every non-constant polynomial eventually runs off to $\pm\infty$ while $\cos\theta$ stays trapped in $[-1, 1]$. The phrase "for all real θ" is what makes the statement false.

Part (iv): true

For $x > 0$,

$$x^4 + 3 + x^{-4} - 5 = x^4 - 2 + x^{-4} = \left(x^2 - x^{-2}\right)^2 \geq 0,$$

since a square of a real number is non-negative. Hence $x^4 + 3 + x^{-4} \geq 5$ for all $x > 0$, with equality exactly when $x^2 = x^{-2}$, that is, when $x = 1$. (Alternatively, the AM–GM inequality gives $x^4 + x^{-4} \geq 2\sqrt{x^4 \cdot x^{-4}} = 2$.)

Summary: (i) true, (ii) false, (iii) false, (iv) true.

Checks

- (i): with $a = 2$, $b = 3$, both $2^{\ln 3}$ and $3^{\ln 2}$ equal about 2.141; random positive a, b agreed to about 10^{-15} relative error.
- (ii): sampling θ over a full period, $\cos(\sin\theta) - \sin(\cos\theta)$ never fell below about 0.107.
- (iii): the degree-20 Maclaurin polynomial for cos is accurate at $\theta = 1$, but is out by about 7.5 at $\theta = 10$ and by about 10^{11} at $\theta = 30$.
- (iv): on a fine grid the minimum of $x^4 + 3 + x^{-4}$ is 5, at $x = 1$.

Common slips

- Trying to "prove" a false statement by checking a few values, or "disproving" a true one by an algebra slip: decide which way each part goes before writing.
- In (iii), forgetting the constant polynomials: the "tends to infinity" argument needs degree at least 1.
- In (iii), quoting the Taylor series as a polynomial that works everywhere; it doesn't.

Beyond STEP

Part (iii) contrasts with the Weierstrass approximation theorem: on any *closed bounded* interval, every continuous function can be approximated uniformly by polynomials to any accuracy you like. Uniform approximation on the whole real line is impossible, exactly because of the growth argument above. Abbott's *Understanding Analysis* discusses the Weierstrass theorem and uniform convergence.

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